

19 (a) (b)	$\frac{1}{18}(8k\mathbf{b} + 10\mathbf{b} - \mathbf{a})$ No, with explanation	P1 for a correct expression for $\overrightarrow{CB}$ or $\overrightarrow{BC}$ eg, $\overrightarrow{CB} = -k\mathbf{b} - \mathbf{a} + \mathbf{b}$ or $\overrightarrow{BC} = -\mathbf{b} + \mathbf{a} + k\mathbf{b}$ P1 for a correct expression for $\overrightarrow{CN}$ or $\overrightarrow{BN}$ or $\overrightarrow{NC}$ or $\overrightarrow{NB}$ eg, $\overrightarrow{CN} = \frac{5}{9}(-k\mathbf{b} - \mathbf{a} + \mathbf{b})$ or $\overrightarrow{BN} = \frac{4}{9}(-\mathbf{b} + \mathbf{a} + k\mathbf{b})$ or $\overrightarrow{NC} = \frac{5}{9}(-\mathbf{b} + \mathbf{a} + k\mathbf{b})$ or $\overrightarrow{NB} = \frac{4}{9}(-k\mathbf{b} - \mathbf{a} + \mathbf{b})$ P1 for a correct unsimplified expression for $\overrightarrow{MN}$ eg $\frac{1}{2}\mathbf{a} + k\mathbf{b} + \frac{5}{9}(-k\mathbf{b} - \mathbf{a} + \mathbf{b})$ oe or $-\frac{1}{2}\mathbf{a} + \mathbf{b} + \frac{4}{9}(-\mathbf{b} + \mathbf{a} + k\mathbf{b})$ oe A1 for $\frac{1}{18}(8k\mathbf{b} + 10\mathbf{b} - \mathbf{a})$ oe eg $\frac{5}{9}\mathbf{b} + \frac{4}{9}k\mathbf{b} - \frac{1}{18}\mathbf{a}$ C1 No with supporting reason ft (a) Acceptable reasons: No, since $\frac{1}{18}(8k\mathbf{b} + 10\mathbf{b} - \mathbf{a})$ is not a multiple of $\mathbf{b}$ No, as N is not the midpoint of BC No, they are not multiples of each other No, does not factorise to $\mathbf{b}$ Not a multiple of OB OB doesn't have an $\mathbf{a}$ Not acceptable reasons: Yes... No, they don't share the same multiples OB doesn't go into MN	All vectors must be clearly identified This mark implies the previous one Must have a maximum of 3 vector terms, $\mathbf{a}$, $\mathbf{b}$, and $k\mathbf{b}$
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